

Telecom Customer Churn Prediction

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3,333

Customers Analysed

14.5%

Annual Churn Rate

5 Models

Tested & Compared

Business Understanding

Setting the stage – why churn matters and what we set out to solve

The Opportunity



Acquiring a new customer costs 5-7× more than retaining one



Even 1% churn reduction = meaningful recurring revenue gain



Proactive outreach is far cheaper than win-back campaigns

The Challenge



14.5% of customers leave each year (~483 annually)



We only know a customer churned after it's too late



No system exists to identify at-risk customers early

Our Goal



Build an AI model to score every customer weekly



Flag high-risk customers before they decide to leave



Give the retention team a prioritised call list

Data Understanding

3,333 customers · 21 features · one clear target: did they leave?

3,333

Total Customers

483

Churned (14.5%)

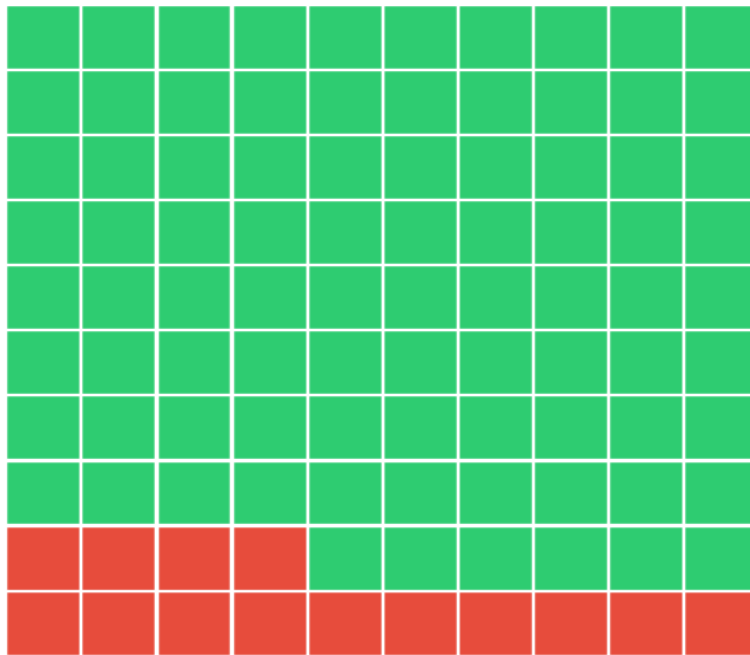
21

Features Analysed

0

Missing Values

Customer Churn Rate — Each Square = 1% of Customers

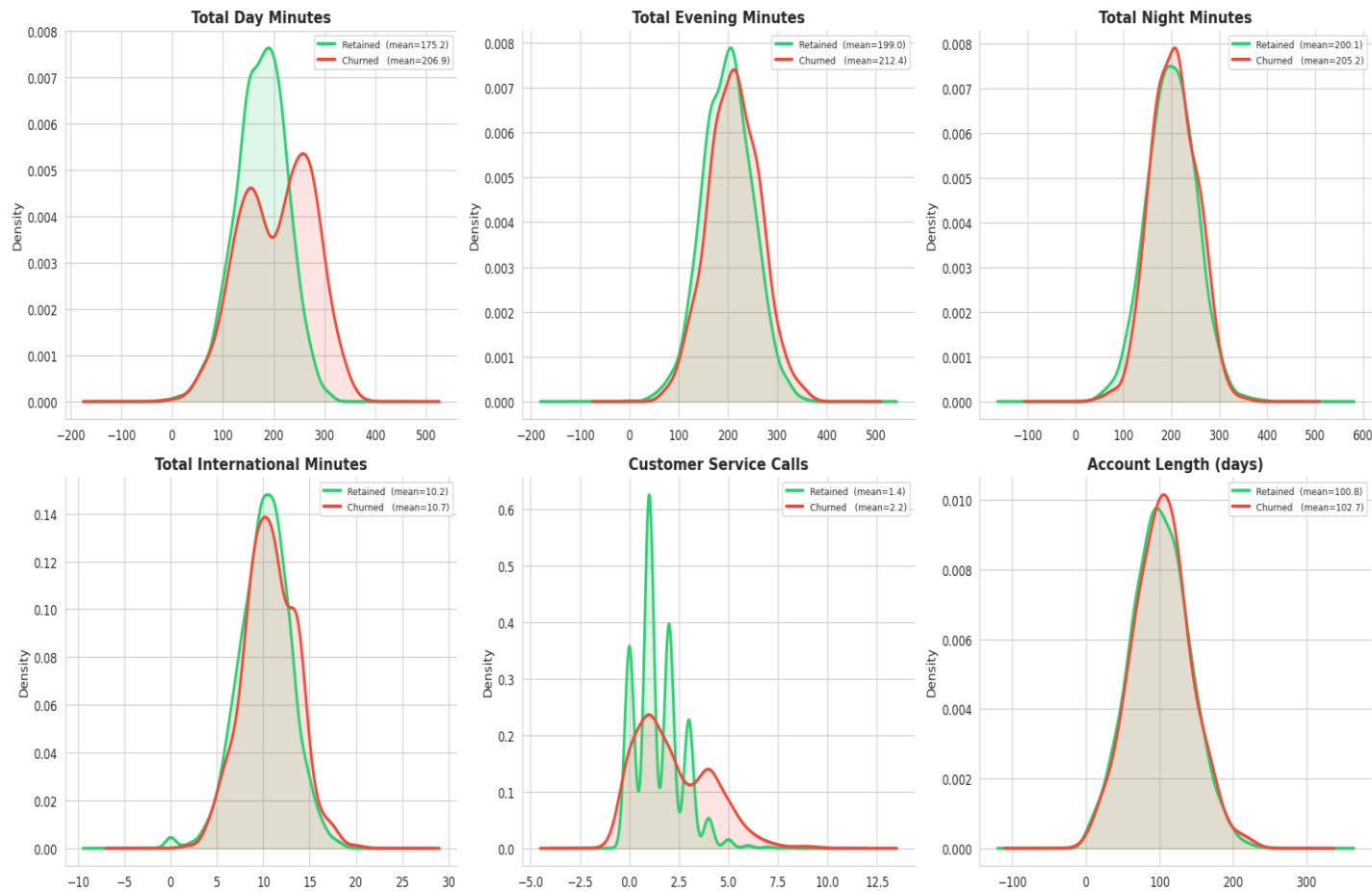


Churned — 483 customers (14.5%)



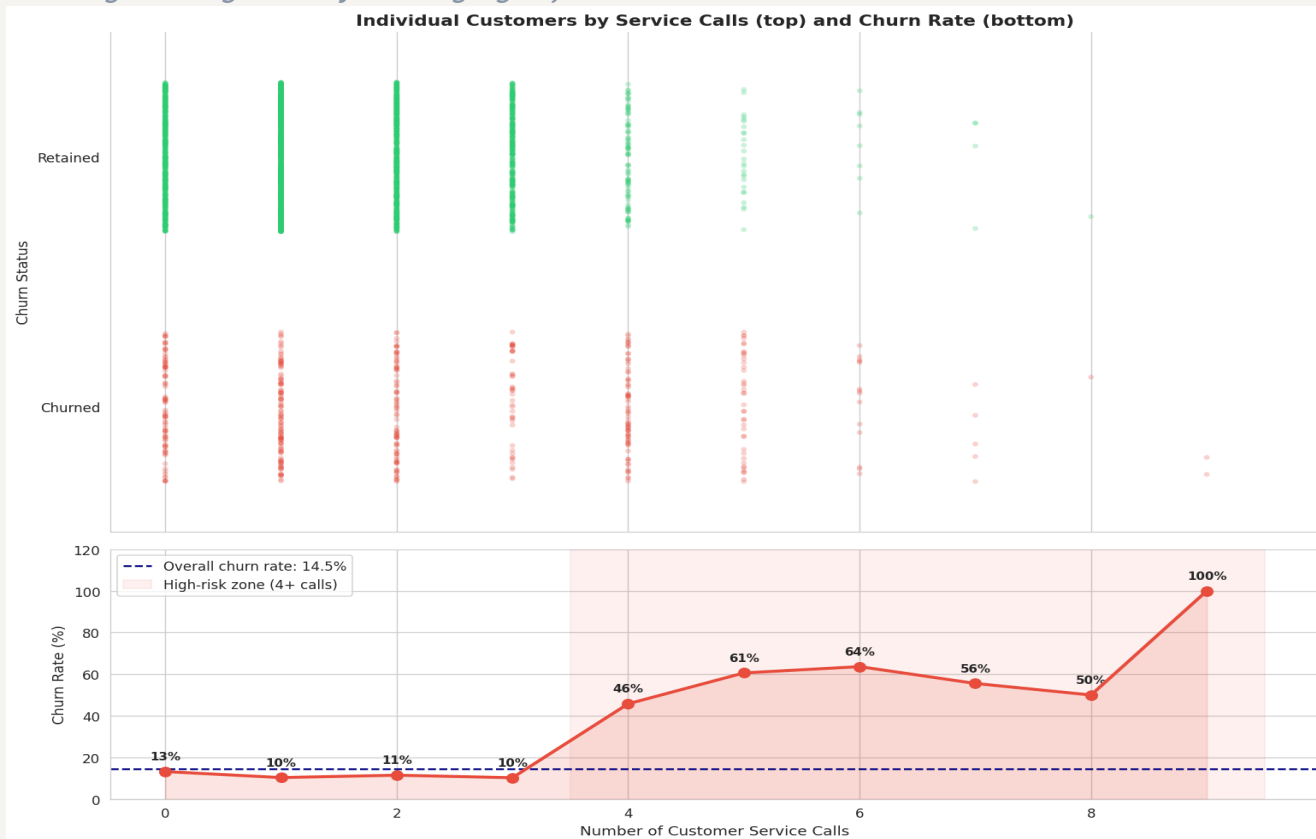
Retained — 2,850 customers (85.5%)

Feature Distributions: Churned vs Retained Customers (KDE)



Data Finding – The Service Call Signal

The single strongest early warning sign of a customer about to leave



Key Insight

~11%

Churn rate for customers with 0-3 service calls

46-100%

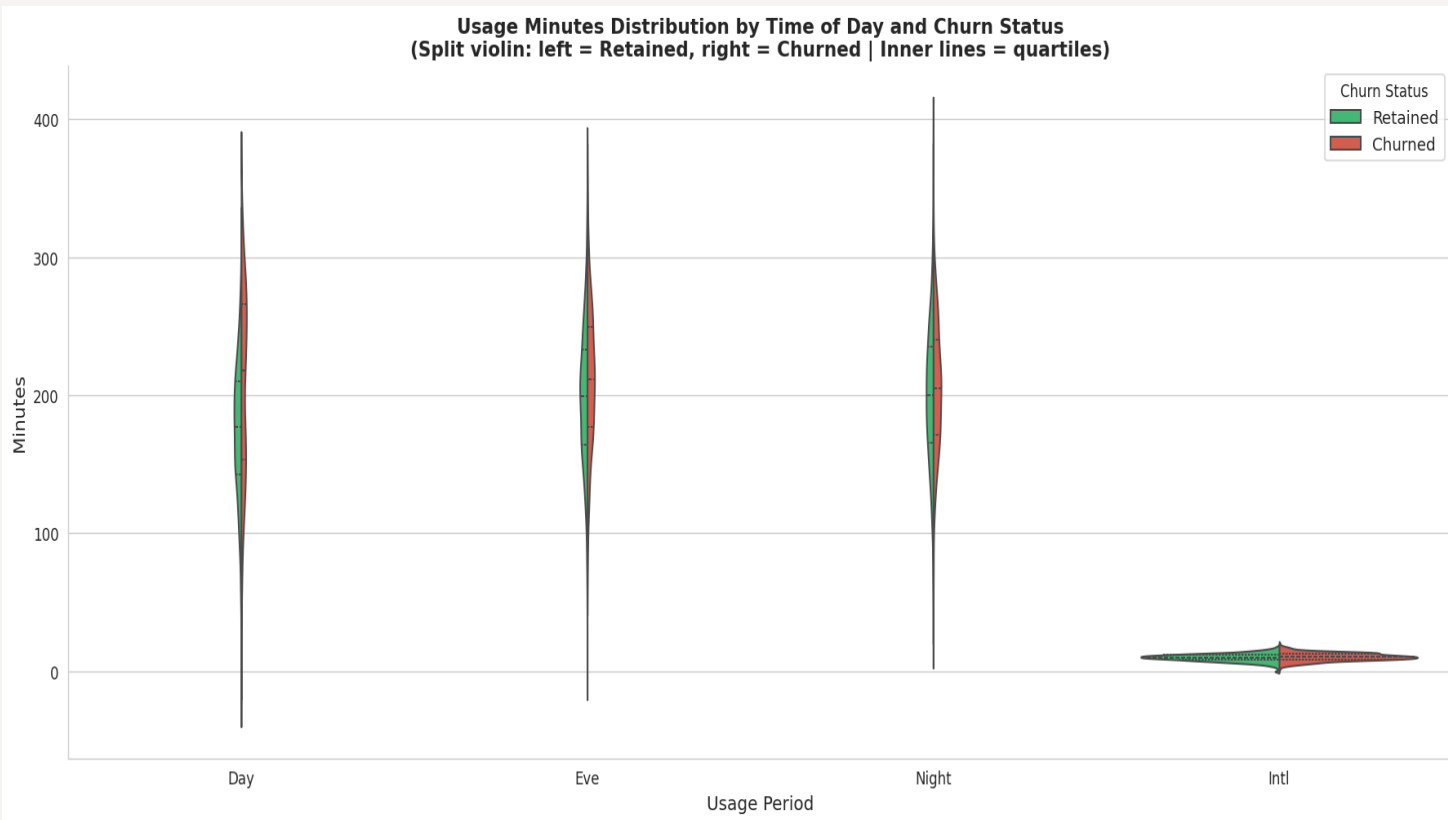
Churn rate for customers with 4+ service calls

Business Implication

After a customer's 3rd call, route them immediately to a senior retention agent – before they decide to leave.

Data Finding – Usage Patterns Drive Churn

Heavy daytime callers face higher bills – and are more likely to leave



Day Minutes

Churners avg **207 min/day** vs 175 for retained – higher usage = higher bills = price pressure

Night & Evening

Night and evening minutes show very similar distributions between churned and retained – these are weaker predictors.

Modeling Approach – Five Models Tested

We tried increasingly powerful approaches until we found the best one

M1

**Logistic Regression
(Baseline)**

Simple but only 25%
recall – missed most
churners

M2

**Logistic Regression
(Tuned)**

Better: 74% recall, but
poor at precision

M3

**Decision Tree
(Baseline)**

Memorised training data
– overfit badly

M4

**Decision Tree
(Tuned)**

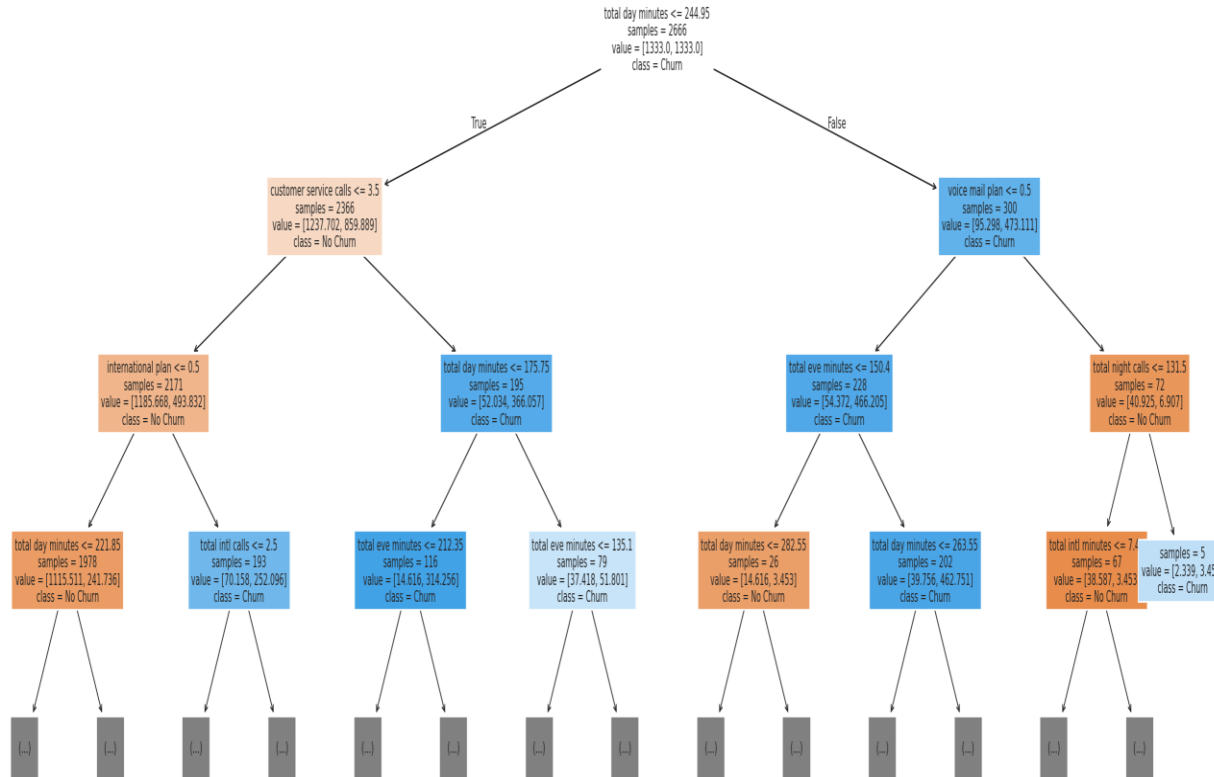
74% recall, better
balance – but still
overfit

M5

**Random Forest
★ FINAL**

Best generalisation,
highest AUC (0.88), most
reliable

Tuned Decision Tree – Top 3 Levels

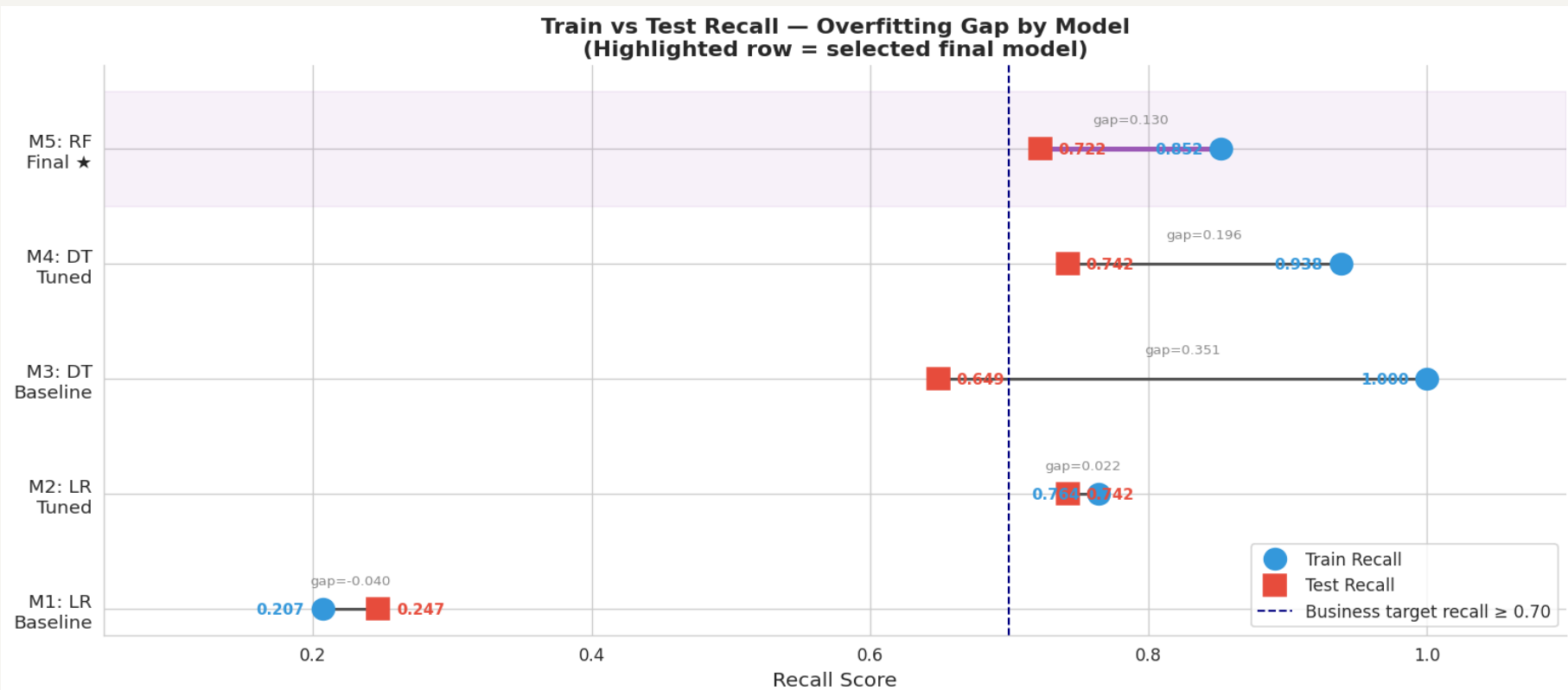


Interpretation
Overfitting gap shrinks from 0.351 to 0.196.
Test recall rises to 74.2% matching M2, but F1 is better (0.673 vs 0.474) — same recall with far less collateral. The tree first splits on customer service calls and total day minutes, confirming the EDA insights.

Model Evaluation — Comparing All Five Models

The Random Forest wins on the metrics that matter most for business

Train vs Test Recall — How Much Does Each Model Overfit?

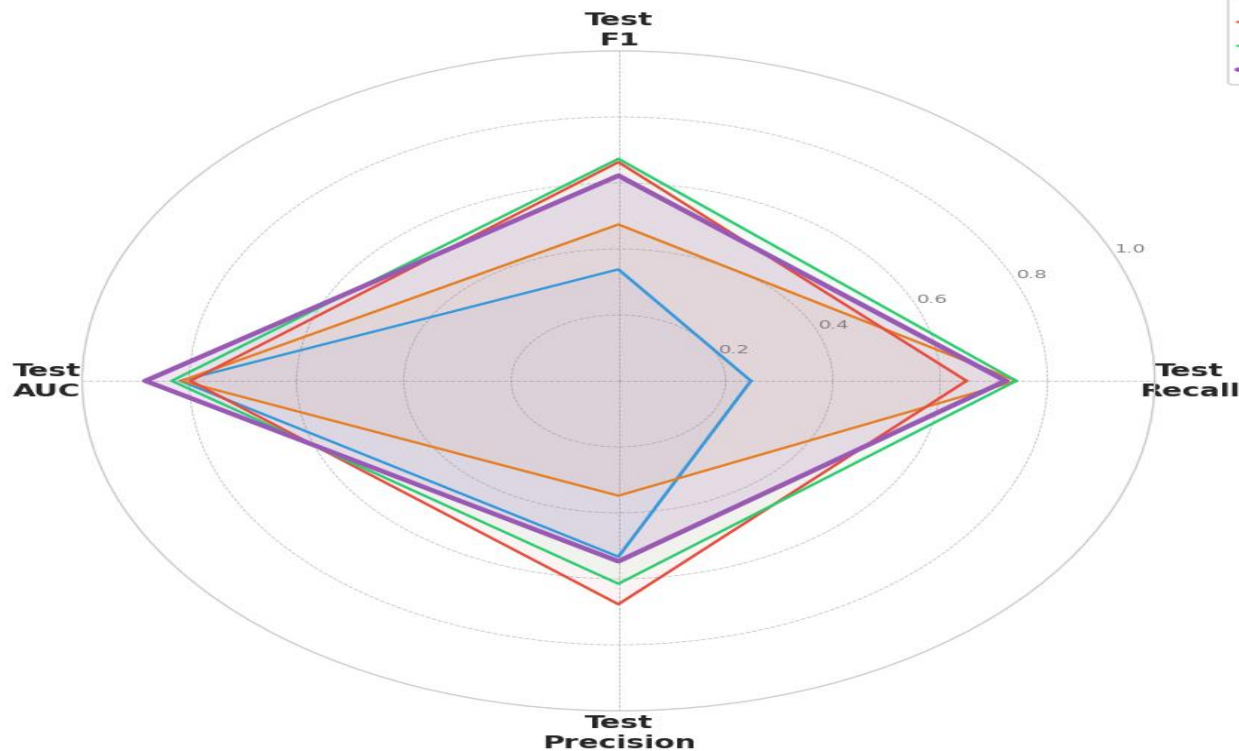


Model Evaluation – Comparing All Five Models

The Random Forest wins on the metrics that matter most for business

All Metrics at Once – Random Forest Leads

Model Comparison — Radar Chart
(All metrics on test set)



72%

Recall

0.88

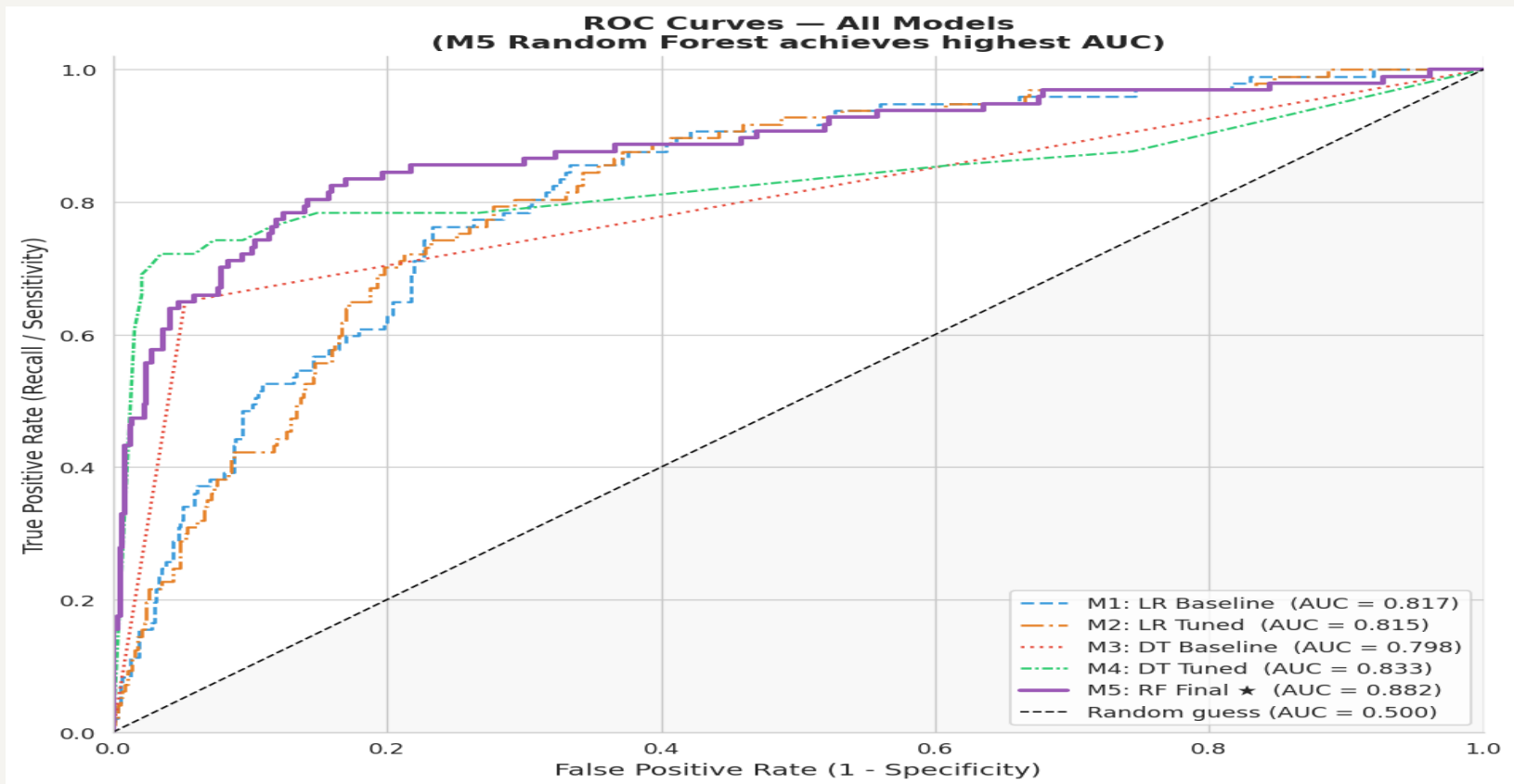
AUC

55%

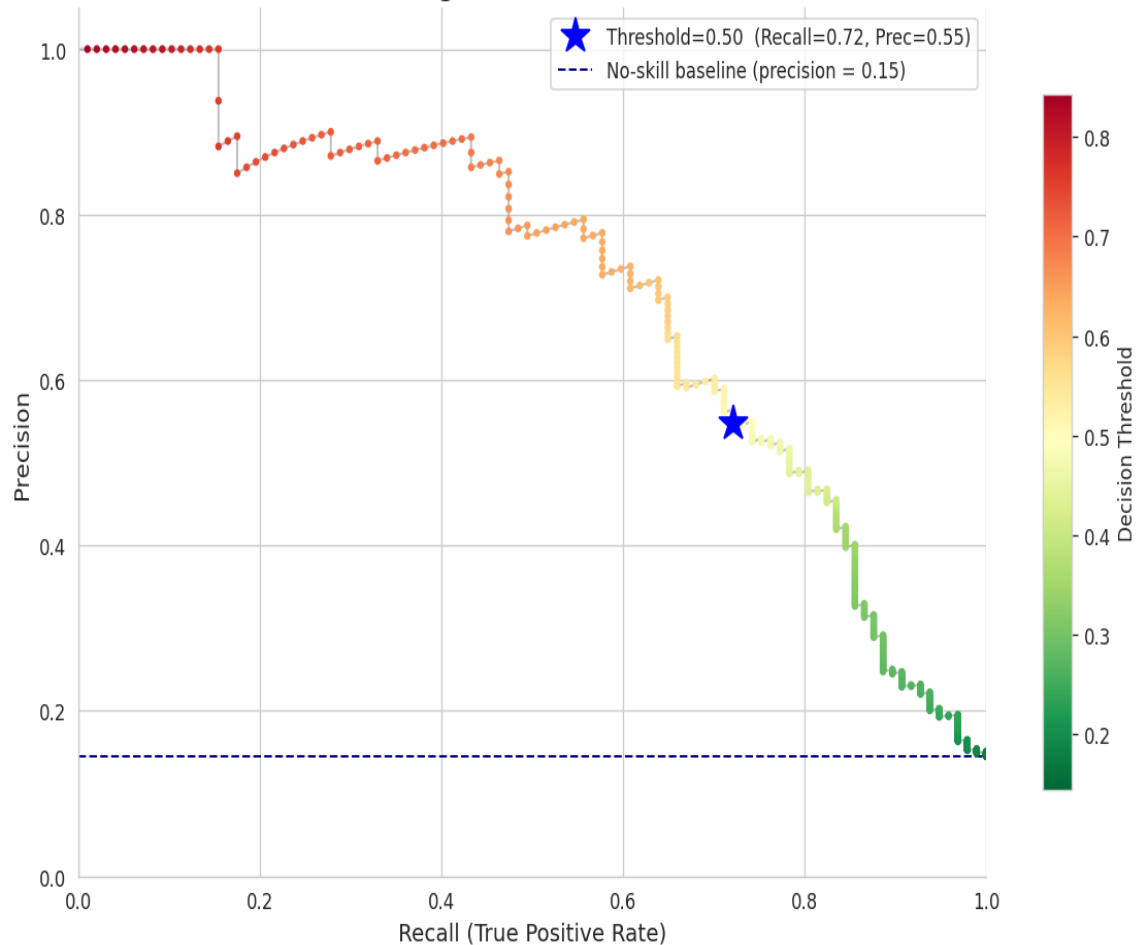
Precision

Model Evaluation – What the Numbers Actually Mean

ROC Curve – All Models



Precision-Recall Curve — Random Forest Final Model
Average Precision = 0.711



Final Model – Confusion Matrix:
How Predictions Break Down

Confusion Matrix
M5: Random Forest (Final)

True label	No Churn	Churn
No Churn	512	58
Churn	27	70
		Predicted label
		No Churn Churn

Model Evaluation — Explanation

Translating model performance into business outcomes

Plain-English Score Summary

Recall 72%

Of every 100 customers who would leave, we catch 72 and can intervene.

Precision 55%

Of every 100 customers we flag, roughly 55 are genuine churners.

AUC 0.88

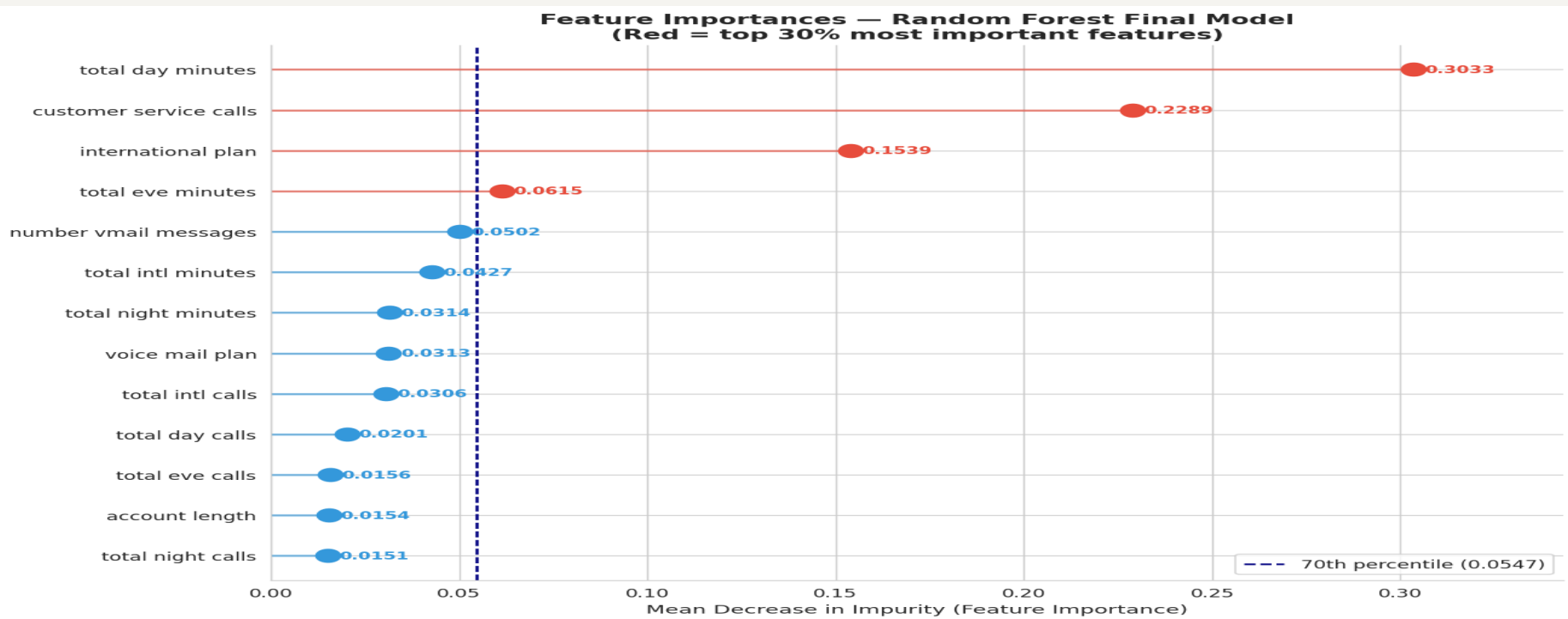
Given any churner and non-churner pair, we rank the churner higher 88% of the time.

28% still missed

We can lower this by adjusting the threshold — trading precision for more recall.

What Drives Churn? – Feature Importance

The model reveals which customer behaviours predict departure most strongly



TOP 3 FEATURES — WHAT THEY MEAN FOR US

Total Day Minutes (0.303)

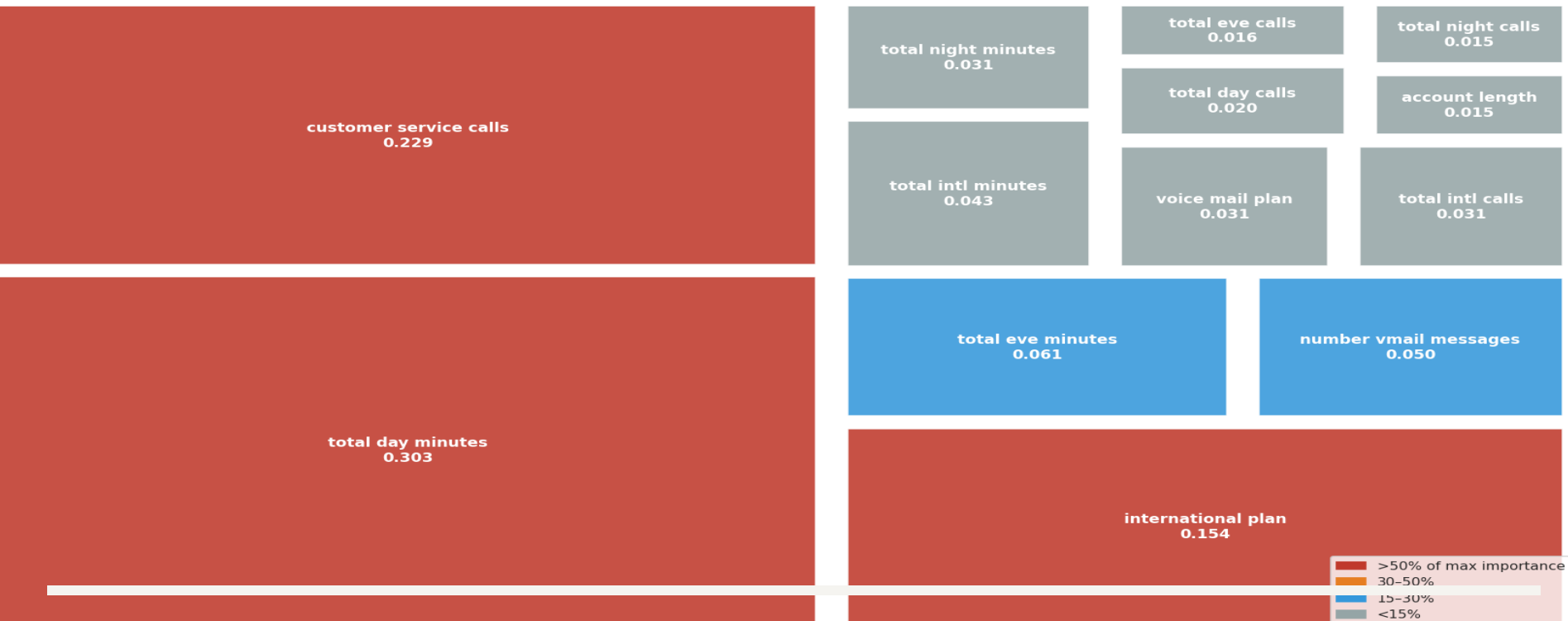
Customer Svc Calls (0.229)

International Plan (0.154)

What Drives Churn? – Feature Importance

The model reveals which customer behaviours predict departure most strongly

Feature Importance Treemap — Random Forest Final Model
Area proportional to importance | Red = highest, grey = lowest



TOP 3 FEATURES – WHAT THEY MEAN FOR US

Total Day Minutes (0.303)

Customer Svc Calls (0.229)

International Plan (0.154)

Our Three Recommendations

Immediate Action

01

Auto-Flag After the 3rd Service Call

Churn rate jumps from ~11% to 46%+ at 4 calls. Route to a senior retention agent before the 4th call happens. Set up a CRM trigger today.

Within 30 Days

02

Rate Cap Loyalty Offer for Heavy Daytime Users

Flag customers above the 75th percentile in day minutes (>216 min/day). Offer a price guarantee — these are your highest-value and highest-risk customers.

Within 60 Days

03

Audit the International Plan + Offer Voicemail as Incentive

42% of intl plan holders churn — review pricing vs competitors. Separately: voicemail plan holders churn at just 8.7% vs 16.7% — offer it free as a retention incentive.

Next Steps – Deployment Roadmap

Moving from model to measurable business impact

Phase 1 Weeks 1-2

- Deploy weekly scoring pipeline to CRM
- Export priority-ranked at-risk customer list
- Set operating threshold with retention team

Phase 2 Month 1-2

- Run A/B test: hold out control group of high-risk customers
- Measure actual revenue retained vs outreach cost
- Tune threshold based on team capacity

Phase 3 Quarterly

- Retrain model on newest data every quarter
- Add contract type & demographic features
- Target: push recall above 80% with richer data

Model limitation: 27.8% of churners are still missed. Richer data and quarterly retraining will close this gap over time.



Thank You

Questions & Discussion

Data Science Team · Moringa School